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Market Research Report for The Lego Group

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Market Context – Augmented Reality Empowered Toys

The global Augmented Reality (AR) gaming market is developing rapidly and is due to reach USD 33.5 billion by 2027, up from USD 3 billion in 2020¹. The recent development of AR-enhanced smartphone applications, 5G networks, AI technologies, and the global COVID-19 pandemic have resulted in increased investment in AR technologies. One of the aims of such investment is to enrich further children's playing experience. Moreover, the UK's toy market shows resilience despite the global pandemic challenges, being this year up by 2% compared to 2019².

Major toy market players such as Mattel and Hasbro have been actively exploring ways of incorporating AR technologies into their toys to allow children interacting with the physical and digital worlds. Hasbro's Hero Vision AR toy³ that lets children pretend to be a superhero and Mattel's AR-activated books⁴ are examples of the successful symbiosis of physical toys and digital technologies.

There are, nevertheless, some challenges associated with creating and promoting the AR-enhanced toys:

- (i) Privacy concerns raised by parents/carers, especially if children are to capture the surrounding environment.
- (ii) The relatively complicated process of setting up AR toys (e.g., creating a user account, connecting to Wi-Fi/Bluetooth) and making it adults' supervision-free.
- (iii) Digital devices (e.g., smartphones) being a physical barrier to playing 'hands-free'⁵.

The Lego Group too, is joining the AR-empowered toy market. Since 2004, the company has transformed successfully into a consistently growing business with a solid digital foundation. This allows for adding AR technology to LEGO's world-famous toy sets of plastic bricks. Additionally, the LEGO Group has a well-established practice of enhancing their products based on customer feedback. The arising question now is

¹ AR Gaming – Global Market Trajectory & Analytics, 2021, https://www.researchandmarkets.com/reports/5028010/ar-gaming-global-market-trajectory-and-analytics?utm_source=dynamic&utm_medium=CI&utm_code=84t2c2&utm_campaign=1412626+-+Global+AR+Gaming+Market+Report+2020+-+Mobile+AR+Gaming+Market+Projected+to+Grow+at+a+41.9%25+CAGR+to+Reach+%2421.7+by+2027&utm_exec=joca220cid

² toyworld, 2022, *NPD says tow market show resilience despite challenging year*, viewed 16 April 2022, <https://toyworldmag.co.uk/npd-says-uk-toy-market-shows-resilience-despite-challenging-year/>

³ HolowatyKrales, E. *Hasbro's new Hero Vision AR toy lets kids pretend to be Iron Man*, 2018, viewed 13 April 2022, <https://www.theverge.com/circuitbreaker/2018/2/12/16990968/iron-man-ar-hasbro-hero-vision-infinity-wars-release>

⁴ Williams, R. *Mattel brings iconic toy characters to life with AR-activated books*, 2020, viewed 13 April 2022, <https://www.marketingdive.com/news/mattel-brings-iconic-toy-characters-to-life-with-ar-activated-books/587331/>

⁵ Vacante, V. *Retrospective: How LEGO Augments Our World*, 2022, viewed 13 April 2022, <https://vrscout.com/news/retrospective-how-lego-augments-our-world/>

whether the strategy of using AR to enhance the play experience will be commercially successful in the long run or not?

In this light, the choice of exploring the UK's digital technologies-based toys market, including AR-empowered toys, is well justified. First, the global COVID pandemic has boosted growth and sales in the UK toy market⁶, providing a good platform for testing new promotional strategies. Second, the majority⁷ of children in the UK by the age of 11 have their own mobile phone, which allows access to the Internet. Third, the UK's e-commerce sector is the most advanced in Europe, with the revenue from online sales being about GBP 693 billion⁸.

Aims of Market Research for The Lego Group

The LEGO Group requested to conduct market research to explore whether the strategy of using AR to improve the playing experience of children will be commercially successful in the long run or not. In other words, collecting and analysing views, experiences, preferences, and opinions on physical and digital technologies-based toys of parents/carers residing in the UK with children aged 4-16 is required.

The aims of the Market Research are the following:

- (i) For children aged 4 - 16 in the UK, measure their preference for physical and digital technologies-based toys, including AR.

In other words, we aim to gain an understanding of why children prefer to play with solely physical toys or choose digital technologies-based toys, including AR-enhanced toys, instead. Then, we strive to investigate whether children have the environment to play with the digital technologies-based toys (e.g., have access to a stable Internet connection and digital devices supporting AR technologies).

Please note that we focus our Research on `digital technologies-based toys` to avoid confusion. We define these toys as physical toys with a playing experience enhanced using a digital device. The AR-empowered toys, in turn, are part of a broader range of digital technologies-based toys.

⁶ Euromonitor International, 2021, *Toys and Games in the United Kingdom*, viewed 13 April 2022, <https://www.euromonitor.com/toys-and-games-in-the-united-kingdom/report>

⁷ The Guardian, 2020, *Most children own mobile phone by age of seven, study finds*, viewed 13 April 2022, <https://www.theguardian.com/society/2020/jan/30/most-children-own-mobile-phone-by-age-of-seven-study-finds>

⁸ Coppola, D. 2022, *E-commerce in the United Kingdom (UK) - statistics & facts*, viewed 13 April 2022, <https://www.statista.com/topics/2333/e-commerce-in-the-united-kingdom/#dossierKeyfigures>

- (ii) Test which marketing messages aimed at parents/carers with children aged 4 – 16 are most likely to be effective.

This part of the Research involves understanding various characteristics of digital technologies-based toys (including AR-enhanced toys) that are important to parents/carers with 4- to 16-year-old children and are likely to influence their decision of buying these toys as opposed to the solely physical ones. The information could then be used to develop marketing campaigns conveying specific messages to buyers.

- (iii) Consider how these messages may have different levels of appeal amongst different types of customers, with a particular focus on parents/carers with children aged 4 – 16.

Here, parents'/carers' attitudes towards digital technologies-based toys (including AR-enhanced toys) and the likelihood of them buying the toys will be assessed to segment the AR-enhanced toy market. The assessment results will be used to build profiles of customers that are 'typical' to a specific segment and further improve marketing campaigns by tailoring them to parents/carers.

- (iv) Measure demand for retail toy stores compared to the existing online toy stores amongst customers and potential customers, focusing on parents/carers with children aged 4 – 16.

This would first include analysing the extent to which various factors would impact parents/carers' decision to shop for digital technologies-based toys (including AR-empowered toys) online or in-store. Next, having such information would help predict if a parent/carer is more likely to shop online or in physical shops. This would help determine whether there would be a need for more retail stores or the online offerings in case of a successful launch of the AR-empowered toys.

Fieldwork Methods

To meet the Research Aims of exploring the UK digital technologies-based toy market, the following three-fold fieldwork will be conducted:

- (i) 30-minute in-home personal interviews⁹ with 60 children aged 4 – 11,
- (ii) Five 60-minute focus groups of eight with children aged 12 – 16,
- (iii) an online sample survey of 1800 parents/carers with 4 – 16-year-old children.

⁹ Boughton, Best Practices: Market Research with Children & Young People, viewed 13 April 2022, <https://info.angelfishfieldwork.com/market-research-fieldwork-blog/best-practices-qual-mr-children-young-people>

In-home personal interviews and **focus groups** with children aged 4-16 will let us collect qualitative insights into the reasons behind children's preference for physical or digital technologies-based toys. This information would complement the findings of the online sample survey of 1800 respondents' (parents/carers of 4–16-year-old children) views on similar matters, which will be recorded in a more quantitative way.

In sum, through the qualitative and quantitative information obtained, the Research Aims of the study will be addressed, and directions for further research set (please refer to the **Proposed Further Research Section**).

In-home personal interviews (as opposed to, say, interviews in toy stores) with the younger children aged 4 – 11 are likely to allow for:

- (i) Children's cooperation as they are likely to feel more comfortable in their own home environment, and interviewers will be able to approach each child individually.
- (ii) Parents'/carers' cooperation as the interview process will be more convenient for them: at home and together with their children.
- (iii) Demonstrating some sample toys to boost children's interest and make the interview process more engaging and easier to understand.
- (iv) Learning first-hand about which toys and how are used in children's everyday life after being bought.

Focus groups with the older children aged 12 – 16 will allow for:

- (i) Improved rapport as children of this age are more likely to open up when surrounded by their peers.
- (ii) Parents'/carers' cooperation as they will be able to observe the interview process, ensuring children's rights are respected.
- (iii) Testing some sample toys by children and receiving their feedback instantly, including generating ideas.

The online sample survey of 1800 respondents (parents or carers resident in the UK who have children aged 4 – 16) will also include parents/carers who have purchased any LEGO product within the last 12 months.

The main advantages of the online data collection method are speed, flexibility and lower costs while maintaining high-quality standards. The survey links can be quickly and easily sent to the respondents'

emails. The participants can then take their time to fill in the Questionnaire at times convenient to them. This is particularly important given that parents/carers are likely to be short on time as they are busy looking after their children. Not rushing will allow the respondents to provide more thoughtful, sincere, and accurate answers.

Combining the three above mentioned data collection methods allows mitigating the drawbacks of each individual method and ensuring a sufficient amount of both qualitative and quantitative information is obtained. **The draft survey Questionnaire is annexed to the Report.** Having made any content and format-related changes after pilot testing the Questionnaire, the exterior design will be optimised to make it aesthetically appealing.

Sampling Methods

Two samples will need to be selected:

- (i) 100 children (60 children aged 4 – 11 and 40 children aged 12 – 16),
- (ii) 1800 parents/carers with 4 – 16-year-old children.

The first sample – 100 children – will be selected from DRG Global's Access Panel¹⁰ of 77,000 children and teens aged 4 – 19 across the UK. The children will be pre-screened to ensure that their parents/carers reside in the UK for consistency with LEGO's research population requirements. Please note the sample size of 100 (60 children aged 4 – 11 and 40 children aged 12 – 16) is chosen to ensure a fair and proportionate representation of children from the two age brackets. Please see **Formula 1 in Technical Appendix** for more details.

A quick **simple random sample selection method** will be used to choose the seventy 4–11-year-olds. It is expected that the information obtained from these children will be of a highly unstructured and qualitative nature, and the primary purpose of it is to fill in the gaps uncovered by the online sample survey of parents/carers. Hence, obtaining a representative sample is not among the top priorities for this part of the research, and the **simple random sample** selection method would suffice.

The thirty 12–16-year-olds will be selected through a **stratified sample** to ensure the children come from roughly similar backgrounds. The children's parents'/carers' income bracket could be a potential stratification factor. This will allow for grouping children from similar backgrounds and conducting more effective focus groups.

¹⁰ DRG Global, *Kids & Teens*, viewed 13 April 2022, <https://drg.global/divisions/panelbase/panels/kids-teens/>

The second sample – 1800 parents/carers with children aged 4 – 16 – will be selected from DRG Global's Access Panel of 350,000 adults aged over sixteen¹¹. These adults will be pre-screened to choose parents/carers with children aged 4-16. **Stratified sampling** will then be applied rather than other more straightforward techniques like random or systematic sampling. This technique will allow for more representative, accurate and precise results.

Stratified sampling will be carried out as follows:

- (i) Parents and carers of interest will be divided into non-overlapping groups by the region they reside in, their social grade and whether they have purchased a LEGO product in the past 12 months. A total of 12 regions¹² from England, Scotland, Wales, and Northern Ireland and 6 socio-economic grades¹³ will be considered. Such groups choice not only allows for LEGO and non-LEGO customers but also for parents/carers from different socio-economic backgrounds and geographical locations to be represented.
- (ii) Appropriate simple random sub-samples that add up to the total sample size will be selected based on the precision required in each of the non-overlapping 144 groups of parents/carers of children aged from 4 to 16 (please see **Formula 2 in Technical Appendix**).

Please note that to complete the above stratification, the following information on the parents/carers of interest will be obtained from the DRG Global's respondents' profiles:

- (i) county residence,
- (ii) socio-economic grade.

To determine whether the potential respondents purchased a LEGO product or not in the past 12 months, their information will be matched to LEGO's customer database in the UK. Such information will be requested from the LEGO Group's Sales Team before starting the fieldwork.

Sampling Frame

The sampling frame – a list of respondents' emails – will be obtained from DRG Global's database. The respondents' socio-demographic information will also be obtained from their database profiles to shorten

¹¹ DRG Global, *Adults*, viewed 13 April 2022, <https://drg.global/divisions/panelbase/panels/kids-teens/>

¹² HM Treasury, 2020, *Country and regional analysis: guidance*, viewed at 14 April 2022, <https://www.gov.uk/government/statistics/country-and-regional-analysis-2020/country-and-regional-analysis-guidance>

¹³ UK geographics, *Social Grade A, B, C1, C2, D, E*, viewed 13 April 2022, <https://www.ukgeographics.co.uk/blog/social-grade-a-b-c1-c2-d-e>

the Questionnaire and reduce the effort by respondents required to fill it in. The respondents' socio-demographic data will later be automatically matched with their responses.

Sample Size and Confidence Intervals

For the online survey, to obtain the required sample size of 1,800 (please refer to **Formula 3 in the Technical Appendix**), the following points were considered:

- (i) The proportion of parents/carers with children aged 4 – 16 likely to complete the Questionnaire,
- (ii) The proportion of parents/carers with children aged 4 – 16 likely to qualify to participate in the online survey,
- (iii) Achieving a 95% level of confidence in our estimates and a 4% tolerance on the sampling error¹⁴, as in a social study.
- (iv) Cost considerations of processing a more considerable amount of respondents' data.

Based on the above points, at the current proposal stage of the Market Research, it is assumed that:

- (i) the average online survey completion rate is 42%, given a Questionnaire of independent 32 questions is employed¹⁵,
- (ii) the incidence rate is 80%.

These two proportions, however, may be updated at the survey pre-test stage as more market information will be obtained.

Regarding the precision of survey results, if a sample proportion of 50% was obtained, then we can be 95% certain that the (0.477, 0.523) interval contains the actual population proportion; please refer to **Formula 4 in the Technical Appendix**.

Information to be Collected

From the fieldwork, we aim to collect three sets of information, including but not limited to:

1. From the in-home personal interviews with 4 – 11-year-olds and focus groups with 12 to 16-year-olds:
 - (i) familiarity of children with digital technologies-based (including AR-empowered) toys,

¹⁴Pollfish, *Margin of Error and Sample Size Calculator*, viewed 13 April 2022, <https://www.pollfish.com/margin-of-error-calculator/>

¹⁵Perzynska, K. 2022, *How to Get Survey Completion Rate of Over 80% (A Real Study)*, viewed at 13 April, 2022, <https://survicate.com/surveys/survey-completion-rate/>

- (ii) whether children have any experience of playing with digital technology-based toys or not, and if yes, what was the experience like,
- (iii) what children like/dislike about digital technology-based toys,
- (iv) whether using digital technologies-based toys independently/with the assistance of adults is accessible or not,
- (v) whether children would like to have their current physical toys possess a digital technologies extension or not.

The qualitative findings from the interviews and focus groups will be used to help understand the reasons behind potential children's preferences toward digital technologies-based toys, including AR-empowered toys.

2. From the online sample survey:

- (i) an estimated proportion of children who have the environment to play with digital technologies-based toys, i.e., children having access to a digital device and stable Internet connection,
- (ii) general characteristics of a digital technologies-based toy that matter to children's parents/carers,
- (iii) factors impacting parents'/carers' decision to buy new digital technologies-based toys,
- (iv) parents'/carers' digital technologies-based toy price sensitivity,
- (v) parents'/carers' shopping preference when buying digital technologies-based toys: in person at physical stores or online.

From the LEGO Group, the following customer and operational data will be collected. The data should preferably be obtained before the start of the fieldwork and supplement the analysis of the fieldwork findings:

- (i) data on LEGO's UK customers who purchased their products in the past 12 months,
- (ii) LEGO customers' socio-demographic profiles,
- (iii) Characteristics of LEGO's existing AR-empowered toys and the extent to which these can be altered. This includes but is not limited to the possibility of changing the design of the physical toys/interface of the digital aspect of the toy, altering age suitability, and introducing extra safety measures,
- (iv) current most popular toys (both physical and AR-empowered) and their price ranges.

Proposed Statistical Analysis: Logistic Regression and Cluster Analysis

Once we complete the fieldwork, we will proceed to analyse collected data/information using two multivariate analysis statistical techniques:

1. **Cluster Analysis** and
2. **Logistic Regression**

Please note that data for the Model will be obtained from the survey results, as all questions have been carefully designed to be used for the statistical analysis. Further note that excessive technical details are omitted in the current Report and may be presented at a later stage of the research.

Cluster Analysis

Cluster Analysis will be applied to understand the parents'/carers' attitudes and preferences regarding the digital technologies-based toy characteristics that matter to them the most. Having this information will allow us to determine:

- (i) what marketing messages are likely to reciprocate with parents/carers,
- (ii) tailor the marketing messages to specific groups of parents/carers to prioritise slightly different digital technologies-based toy characteristics and influence their buying decisions.

In other words, Cluster Analysis will aid segmenting the digital technologies-based toy market to understand the profiles of customers/potential customers. This will help tailor LEGO's products' marketing campaigns, determine toy pricing strategies, and offer competitive deals to help outcompete other toy companies. For instance, the LEGO Group could define their price ranges based on whether the customers shop online or in-person or if specific sets of digital technologies-based toys are more appealing to certain customers.

Based on a pre-selected set of characteristics like the toy's safety, affordability or whether it contains an educational element or not, an appropriate number of sensible clusters will be determined and then profiled based on the customers' social and geo-demographic and economic characteristics, such as age group, residence area, gender, number of children and their gender, education and income level.

Employing this statistical analysis technique is particularly useful in the absence of past research on parents/carers market segmentation regarding their views on toy characteristics. Indeed, there is no a-priori information on which segments parents/carers could belong to, allowing for the technique to detect naturally occurring and hence most practical segments.

Logistic Regression

Logistic Regression will aid in estimating demand for retail toy stores compared to the demand for the online toy stores amongst customers and potential customers (i.e., parents/carers with children aged 4 – 16) for buying digital technologies-based toys. The Logistic Regression is easily understandable and interpretable, yet it allows for making sophisticated conclusions.

A statistical Model will be built to predict the probability of a given person choosing to shop in-person or online. Given a certain amount of information on a group of customers/potential customers, the Model will help determine what proportion of customers is likely to shop in physical stores or order the digital technologies-based toys online.

More precisely, having such a Model will allow us to:

- (i) estimate the proportion of customers/potential customers likely to shop for digital technologies-based toys in-person or online,
- (ii) build a profile of the 'ideal' parent/carer who is likely to shop for digital technologies-based toys online or in person. The profile will be specified in terms of the parent's/carer's socio- and geo-demographic characteristics and buying attitudes and behaviours.

Characteristics that will be used to predict the probability of a person shopping for a digital technologies-based toy in-person or online include but are not limited to:

- (i) parents'/carers' socio and geo-demographic characteristics such as age group, residence area, number of children and their gender, educational level and income status,
- (ii) parents'/carers' toy buying attitude and behaviour such as ways of choosing a toy, reasons for shopping online and in-person and types of toy stores preferred.

Please note that the qualitative information specified above will be coded to allow for it to be used in the statistical analysis.

Proposed Further Research

To assist the LEGO Group in pursuing its growth objective, complementary research may be needed to explore the Research Aims in more detail.

Prior to the current research, an Exploratory Study could be deployed to gain a more in-depth understanding of the characteristics of digital technologies-based toys (including AR-empowered) that

parents/carers value. A series of in-depth interviews with toy industry experts, family psychologists, education and IT specialists could be carried out. The preliminary findings could then be used to tailor the current research direction and aid in building the Questionnaire for the online sample survey.

In the current research, additional Research Aims may be considered to maximise the LEGO Group's sales. For instance, determining the price elasticity of digital technologies-based toys' demand while accounting for the specifics of the British culture could be looked at. This could assist the LEGO Group set pricing strategies to provide the best 'sales-profit' trade-off. Another potential objective would be learning about adults' interest in LEGO's digital technologies-based toys (including AR-empowered), especially those part of the collectable series.

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Technical Appendix

Formula 1. Determining the number of children from each of the 4-11 and 12-16 age brackets to be included in the sample size of 100:

$$\begin{aligned} \text{Number of 4 – 11 year olds} &= \text{total sample size of 100} * \frac{8}{13} \approx 60 \text{ children} \\ \text{Number of 12 – 16 year olds} &= \text{total sample size of 100} * \frac{5}{13} \approx 40 \text{ children} \end{aligned}$$

Please note that to ensure proportionate representation of children from each age bracket the following was used:

- Total 4-16 age bracket: 13 years
- 4-11 age bracket: 8 years
- 12-16 age bracket: 5 years

Formula 2. Determining the number of groups (strata) to be used for the stratified sampling of 1800 parents/carers:

$$\begin{aligned} \text{Number of groups(strata)} &= 12 \text{ regions} * 6 \text{ socio – economic grades} \\ &* 2 \text{ groups of parents/carers (LEGO customers and non – customers)} = 144 \text{ groupes} \end{aligned}$$

Formula 3. Explaining how the sample size of 1800 parents/carers was obtained:

$$\begin{aligned} \text{Sample size} &= \frac{\left(\frac{z_{\alpha}}{2}\right)^2 * \text{sample proportion} * (1 - \text{sample proportion})}{\text{tolerance on sampling error}^2} \\ &* \frac{1}{\text{completion rate} * \text{incidence rate}} = \frac{1.96^2 * 0.5 * 0.5}{0.04^2} * \frac{1}{0.42 * 0.8} \approx 1,800 \end{aligned}$$

Please note that according to the **Sample Size and Confidence Intervals Section** of the Report the following was assumed:

- $\frac{z_{\alpha}}{2} = 1.96$
- $\text{tolerance on sampling error} = 0.04$
- $\text{completion rate} = 0.42$
- $\text{incidence rate} = 0.8$

Formula 4. Given survey results, calculating the confidence interval for a sample proportion of 50%:

$$\begin{aligned}\text{Confidence interval} &= \\ &= (\text{sample proportion} - z_{\frac{\alpha}{2}} * \sqrt{\frac{\text{sample proportion} * (1 - \text{sample proportion})}{\text{sample size}}}, \\ &\quad \text{sample proportion} + z_{\frac{\alpha}{2}} * \sqrt{\frac{\text{sample proportion} * (1 - \text{sample proportion})}{\text{sample size}}}) = \\ &= (0.5 - 1.96 * \sqrt{\frac{0.5 * 0.5}{1800}}, 0.5 + 1.96 * \sqrt{\frac{0.5 * 0.5}{1800}}) \approx \\ &\quad \approx (0.477, 0.523)\end{aligned}$$

Please note that according to the **Sample Size and Confidence Intervals Section** of the Report the following was assumed:

- $z_{\frac{\alpha}{2}} = 1.96$
- $\text{sample size} = 1800$
- $\text{sample proportion} = 0.5$ (the most conservative choice)